

EMMANUEL NWORIE

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PROFESSIONAL SUMMARY

PhD candidate in Computational and Applied Mathematics specializing in data science, quantitative modeling, and scientific computing. Skilled in Python, R, and MATLAB with experience building predictive models, analyzing complex datasets, and solving PDE-driven systems. Proven ability to translate mathematical theory into practical, data-driven insights for decision-making.

TECHNICAL SKILLS

Programming: Python, R, MATLAB

Data Science: Pandas, NumPy, SciPy, Statistical Modeling, Machine Learning

Modeling: PDEs, Finite Element Methods, Numerical Simulation

Tools: Jupyter, Git, Linux, LaTeX

EXPERIENCE

Graduate Tutor – SMU (2024–Present)

- Delivered quantitative instruction across calculus, statistics, and linear algebra to 100+ students, improving performance and retention.
- Applied data-driven approaches to diagnose learning gaps and optimize student outcomes.

Graduate Teaching Assistant – SMU (2021–2024)

- Led recitations and supported instruction in applied mathematics and statistics courses.
- Built computational examples using Python and MATLAB to enhance conceptual understanding.

PROJECTS

Heart Disease Prediction Model (R)

- Built a classification model using real-world health data to predict heart disease outcomes.
- Performed data cleaning, feature selection, and model evaluation using statistical techniques.
- Achieved strong predictive performance and identified key risk factors through analysis.

Diffusion-Limited Growth Simulation

- Developed finite element models to simulate PDE-based transport and growth processes.
- Implemented numerical solvers and analyzed system behavior through simulation.

EDUCATION

PhD, Computational & Applied Mathematics – SMU (2021–Present)

M.S., Computational & Applied Mathematics – SMU (2023)

B.S., Mathematics (Honors)